11A
MOLECULAR SHAPE

UNIT 04: FROM MICROSCOPIC TO MACROSCOPIC

CHEMISTRY 1310

MIOY T. HUYNH

LEARNING OBJECTIVES

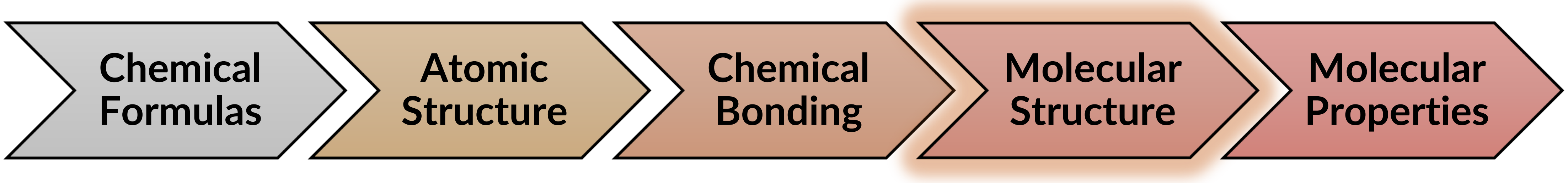
CHAPTER 11

1. Use VSEPR theory and Lewis structures to predict molecular shapes.
2. Identify polar and nonpolar molecules using electronegativity values and molecular shape.
3. Explain valence bond theory.
4. Describe and recognize *sp*, sp^2, and sp^3 hybridization based on the Lewis structure of a molecule.
5. Define, describe, and identify sigma (σ-) and pi (π-) bonds in molecules.

Where did we start and where are we going?

Chemistry in Context

What is our overall goal?



VSEPR and Molecular Geometry

11.1

Valence shell electron pair repulsion (VSEPR) theory

- Used to predict the shape of a molecule based on minimizing repulsion between electron domains around the central atom.
- How to identify and count electron domains?

each single bond	–	one domain
each double bond	=	one domain
each triple bond	≡	one domain
each lone pair	··	one domain

VSEPR and Molecular Geometry

11.1

Electron geometry looks at the arrangement of total *electron domains*.

- Most stable arrangement of electron domains around the central atom (A) because it results in the *least repulsion* between domains.

Molecular shape looks only at how the *atoms* are arranged in the molecule.

VSEPR and Molecular Geometry

11.1

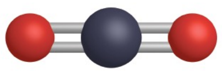
Total number of domains	Electron geometry	Molecular shape
2 domains		?
3 domains		?
4 domains		?
5 domains		?
6 domains		?

There are five electron geometries.

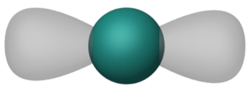
- Based on number of electron domains on the central atom.
- If no lone pairs on central atom:
- If lone pairs on central atom:

VSEPR and Molecular Geometry

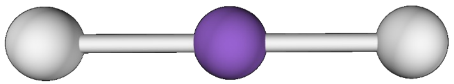
11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
2 domains	0	2	linear	linear	180°	CO_2 

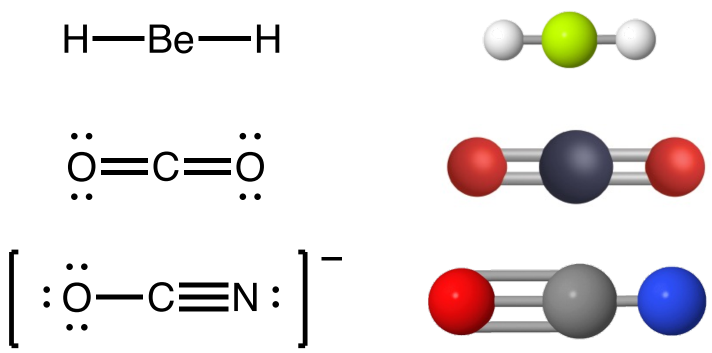
Electron geometry



Molecular shape

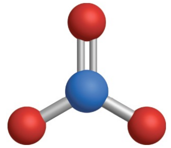
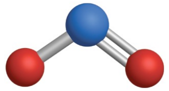


Other examples

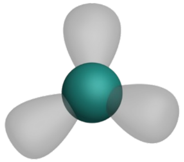


VSEPR and Molecular Geometry

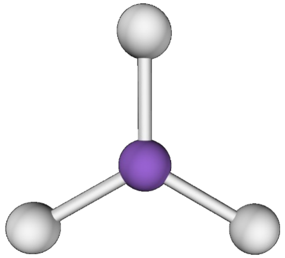
11.1

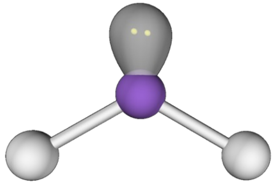
Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
3 domains	0	3	trigonal planar	trigonal planar	120°	NO_3^- 
3 domains	1	2	trigonal planar	bent	< 120°	NO_2^- 

Electron geometry

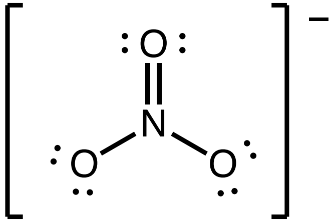


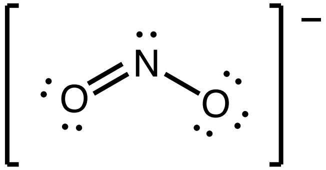
Molecular shapes





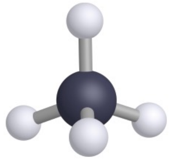
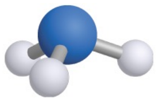
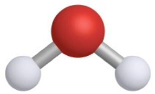
Lewis structures



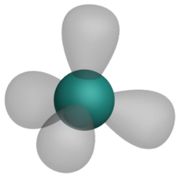


VSEPR and Molecular Geometry

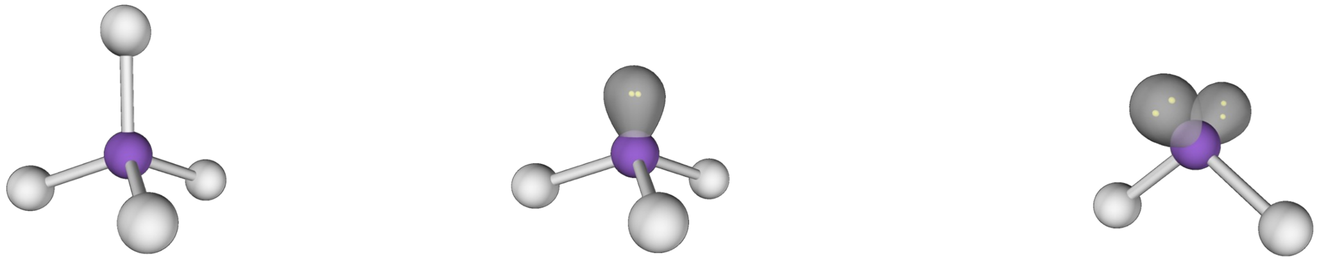
11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model	
4 domains	0	4	tetrahedral	tetrahedral	109.5°	CH_4	
4 domains	1	3	tetrahedral	trigonal pyramidal	< 109.5°	NH_3	
4 domains	2	2	tetrahedral	bent	< 109.5°	H_2O	

Electron geometry

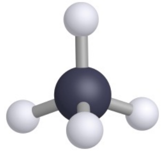


Molecular shapes

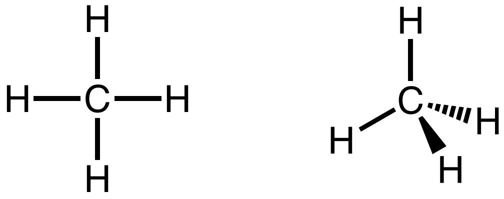


VSEPR and Molecular Geometry

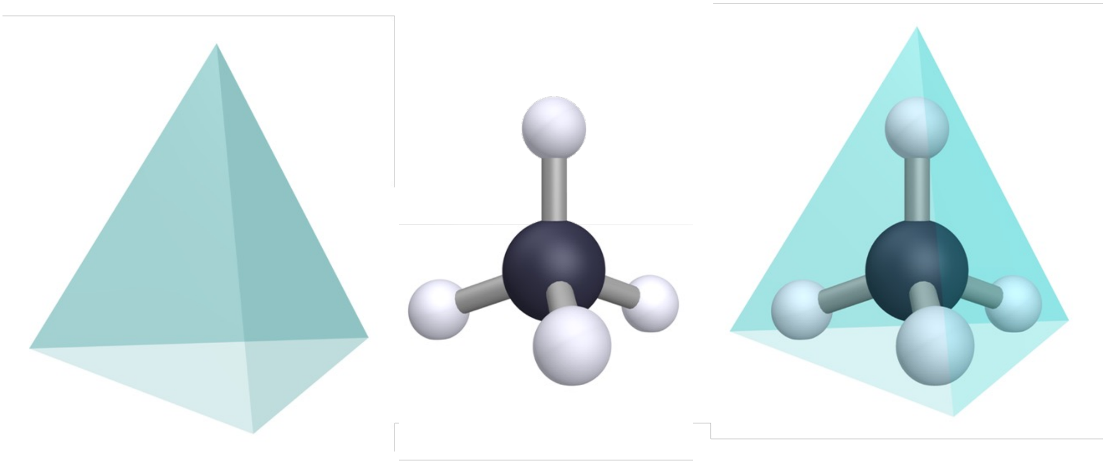
11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
4 domains	0	4	tetrahedral	tetrahedral	109.5°	CH_4 

<u>*2D Lewis structure of methane*</u>

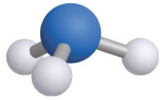


<u>*3D structure of methane*</u>



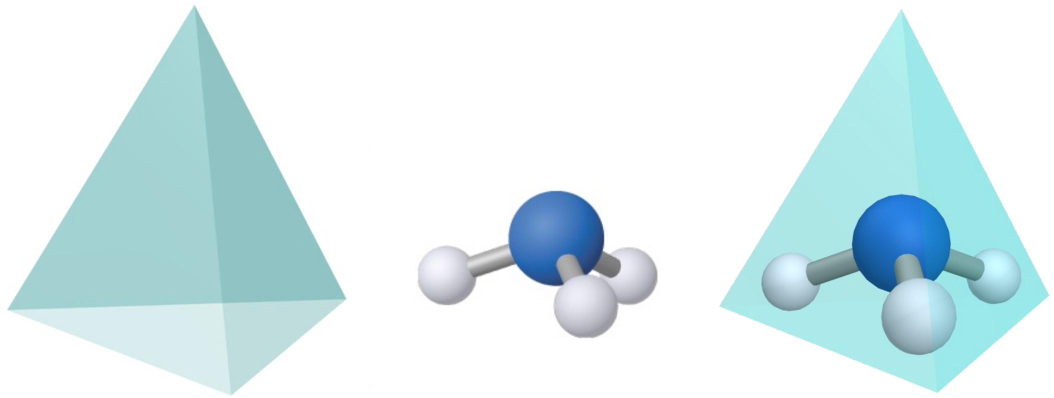
VSEPR and Molecular Geometry

11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
4 domains	1	3	tetrahedral	trigonal pyramidal	< 109.5°	NH_3 

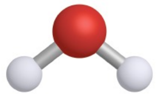
2D Lewis structure of ammonia

3D structure of ammonia



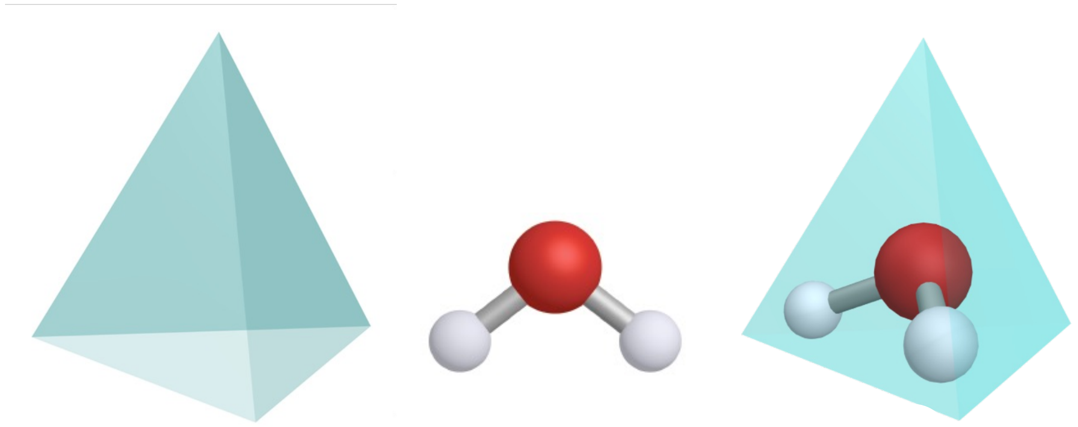
VSEPR and Molecular Geometry

11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
4 domains	2	2	tetrahedral	bent	< 109.5°	H_2O 

2D Lewis structure of water

3D structure of water



IN-CLASS QUESTION 1

Learning Objective(s): 11.1

Draw the Lewis structure for formaldehyde, H_2CO.

Then, complete the table for the molecule to describe the molecular shape.

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape

VSEPR and Molecular Geometry

11.1

Molecules with multiple central atoms

- Any “non-terminal” atom (an atom with only one bonding domain) is considered a “central” atom.
- Shapes of larger molecules are usually described as a series of smaller shapes linked together.

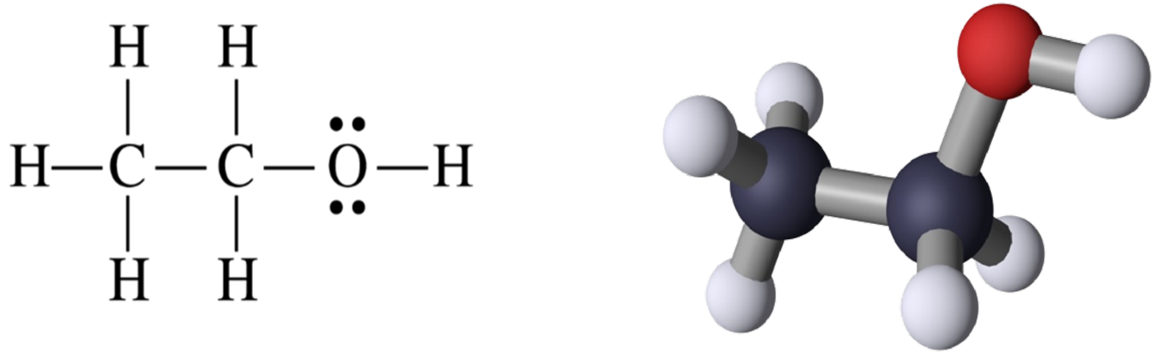
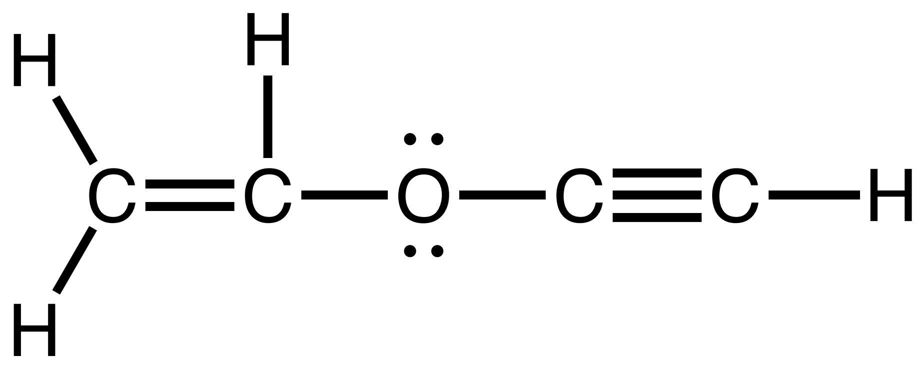


FIGURE 11.10 Lewis structure and 3D model of ethanol (CH_3CH_2OH)

IN-CLASS QUESTION 2

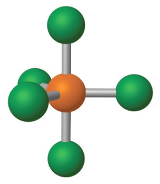
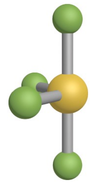
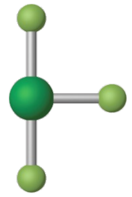
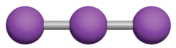
Learning Objective(s): 11.1

Determine the molecular shape for each central atom in the molecule drawn below.



VSEPR and Molecular Geometry

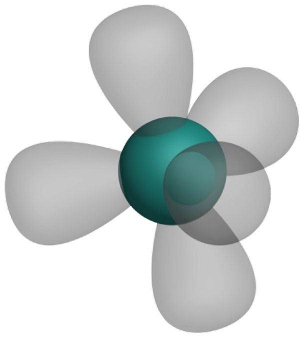
11.1

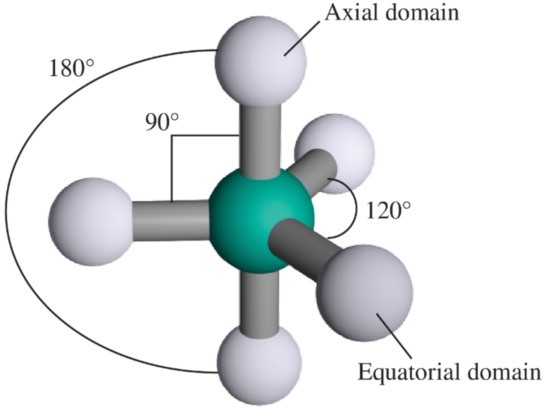
Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model	
5 domains	0	5	trigonal bipyramidal	trigonal bipyramidal	90° 120° 180°	PCl_5	
5 domains	1	4	trigonal bipyramidal	seesaw	< 90° < 120° < 180°	SF_4	
5 domains	2	3	trigonal bipyramidal	T-shaped	< 90° < 180°	ClF_3	
5 domains	3	2	trigonal bipyramidal	linear	180°	I_3^-	

VSEPR and Molecular Geometry

11.1

Electron geometry

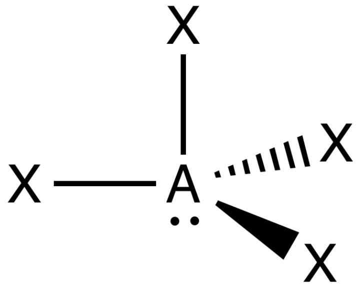




For trigonal bipyramid electron geometry:

Lone pairs occupy equatorial positions!

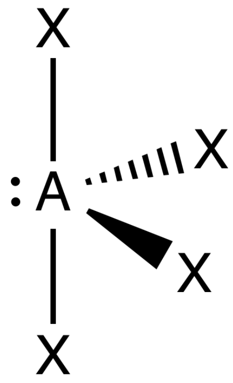
Axial



Three 90° e^- pair repulsions

One 180° e^- pair repulsion

Equatorial

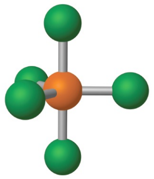


Two 90° e^- pair repulsions

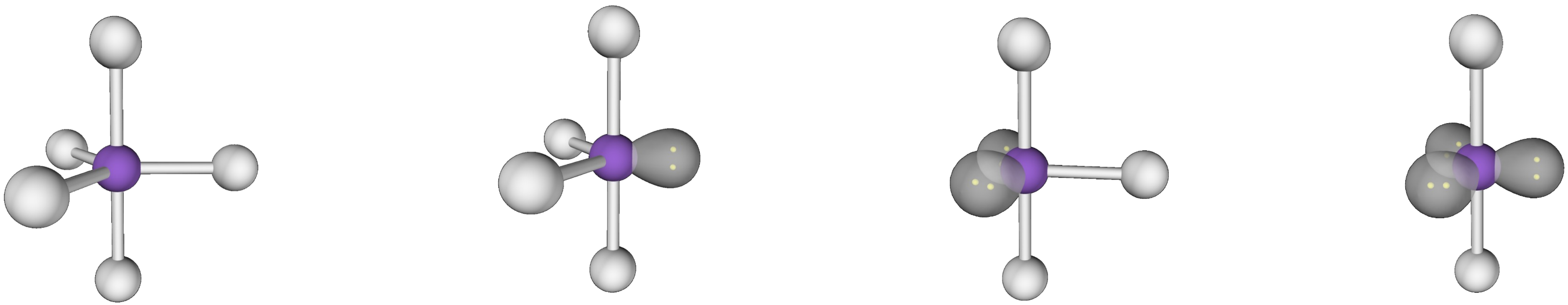
Two 120° e^- pair repulsions

VSEPR and Molecular Geometry

11.1

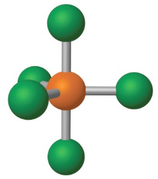
Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
5 domains	0	5	trigonal bipyramidal	trigonal bipyramidal	90° 120° 180°	PCl_5 

Molecular shapes

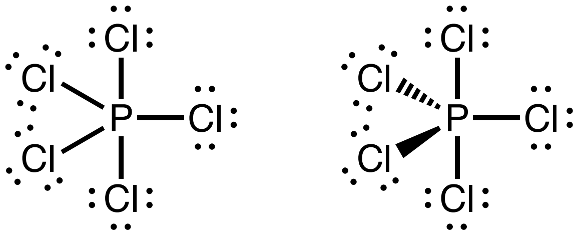


VSEPR and Molecular Geometry

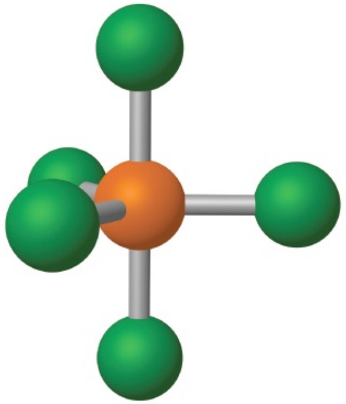
11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
5 domains	0	5	trigonal bipyramidal	trigonal bipyramidal	90° 120° 180°	PCl_5 

2D Lewis structure

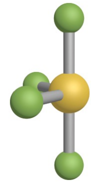


3D structure

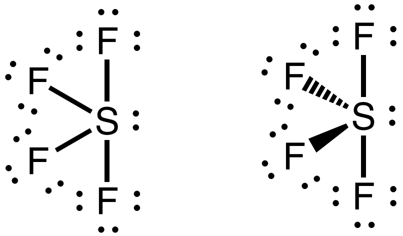


VSEPR and Molecular Geometry

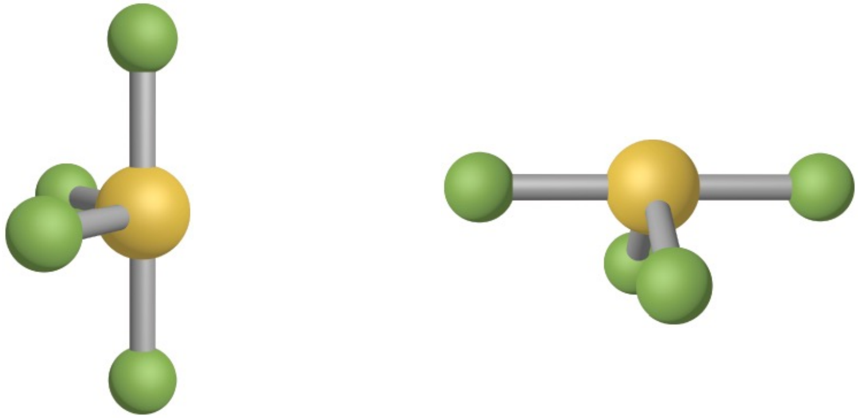
11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
5 domains	1	4	trigonal bipyramidal	seesaw	< 90° < 120° < 180°	SF_4 

2D Lewis structure

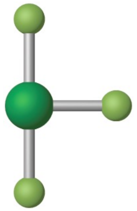


3D structure

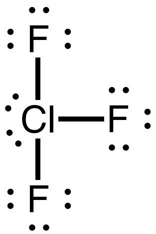


VSEPR and Molecular Geometry

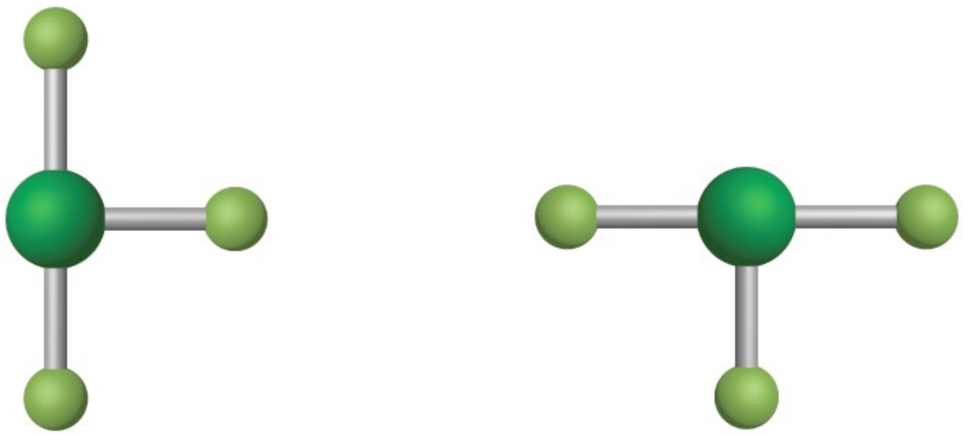
11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
5 domains	2	3	trigonal bipyramidal	T-shaped	< 90° < 180°	ClF_3 

2D Lewis structure

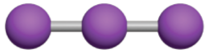


3D structure

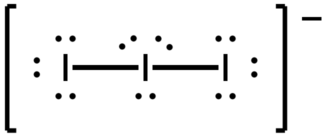


VSEPR and Molecular Geometry

11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
5 domains	3	2	trigonal bipyramidal	linear	180°	I_3^- 

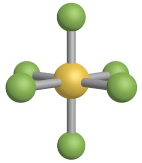
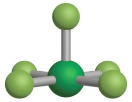
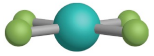
<u>*2D Lewis structure*</u>



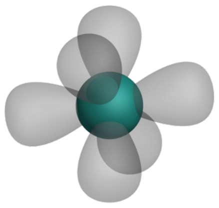
<u>*3D structure*</u>

VSEPR and Molecular Geometry

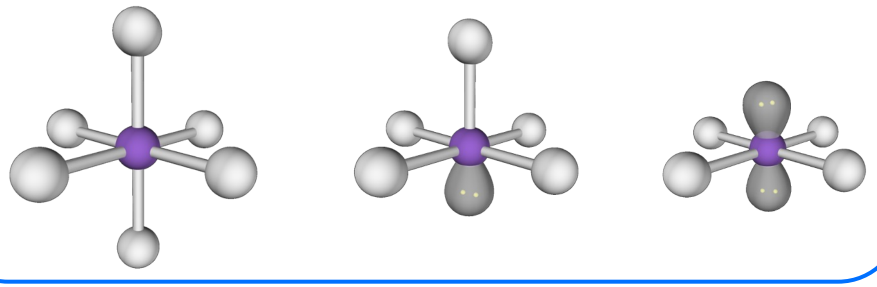
11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
6 domains	0	6	octahedral	octahedral	90° 180°	SF_6 
6 domains	1	5	octahedral	square pyramidal	< 90° < 180°	ClF_5 
6 domains	2	4	octahedral	square planar	90° 180°	XeF_4 

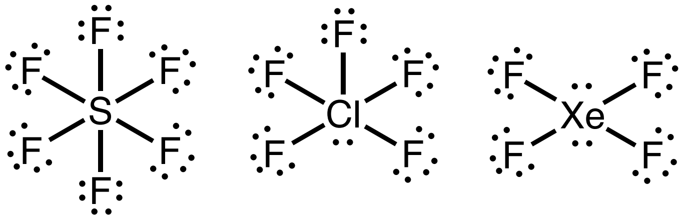
Electron geometry



Molecular shapes



Lewis structures



EXAMPLE PROBLEM

Learning Objective(s): 11.1

Draw the Lewis structures for $SeBr_4$ and BrF_5.

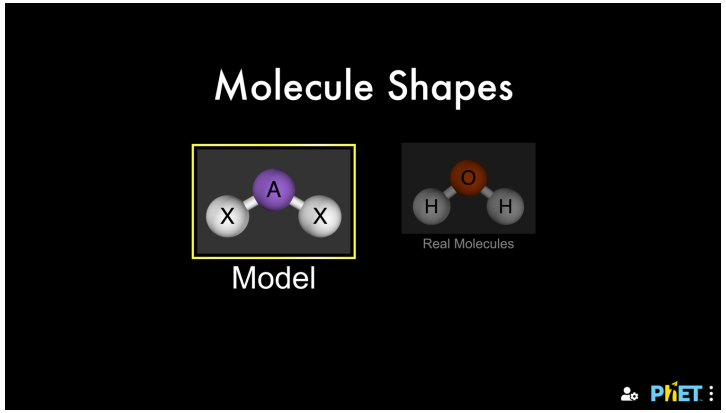
Then, complete the table for the molecules to describe their molecular shapes.

Molecule	Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape
$SeBr_4$					
BrF_5					

Visualizing 3D Molecular Shapes

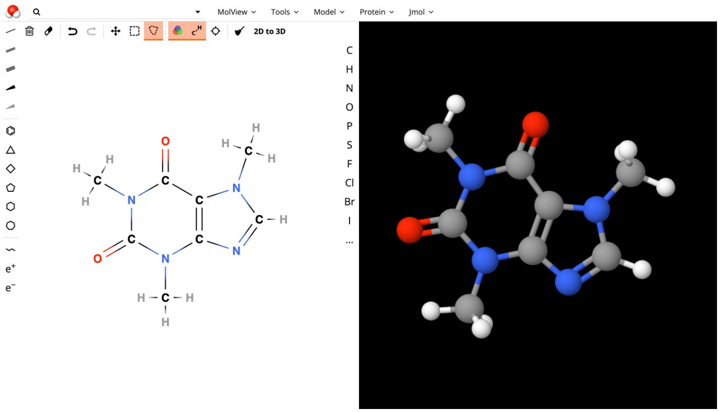
Chemistry in Context

Virtual Tools



https://phet.colorado.edu/sims/html/molecule-shapes/latest/molecule-shapes_all.html

https://molview.org/



Physical Tools

VSEPR model kits

Monday, April 14

BONUS QUESTION

Assessed on: Fall 2021, Exam 3

Identify the molecular shape (also called molecular geometry) for the hypothetical molecule XF_3, where X is an element with 7 valence electrons.

- ❑ Linear
- ❑ Trigonal planar
- ❑ Bent
- ❑ Tetrahedral
- ❑ Trigonal pyramidal
- ❑ Trigonal bipyramidal
- ❑ Seesaw
- ❑ T-shaped
- ❑ Octahedral
- ❑ Square pyramidal
- ❑ Square planar